Uyuni 2026.10

MCP Server Guide



UYUNI

Preface

MCP Server Guide | Uyuni 2026.10

The Uyuni MCP Server connects Model Context Protocol (MCP) clients and AI assistants to the Uyuni API. It provides tools for inspecting systems, assessing patches and CVEs, reviewing scheduled actions, and, when explicitly enabled, changing managed systems.

This guide covers deployment, configuration, client integration, security, available tools, and troubleshooting.

Publication Date: 2026-10-26

Copyright © 2011–2026 Uyuni contributors. All rights reserved. Permission is granted to copy, distribute and/or modify this document under the terms of the GNU Free Documentation License, Version 1.2 or (at your option) version 1.3; with the Invariant Section being this copyright notice and license. A copy of the license version 1.2 is included in the section entitled **Legal Guide › GNU Free Documentation License**.

For SUSE trademarks, see https://www.suse.com/company/legal/ . All third-party trademarks are the property of their respective owners. Trademark symbols (®, ™ etc.) denote trademarks of SUSE and its affiliates. Asterisks (*) denote third-party trademarks. All information found in this book has been compiled with utmost attention to detail. However, this does not guarantee complete accuracy. Neither SUSE LLC, its affiliates, the authors nor the translators shall be held liable for possible errors or the consequences thereof.

차례

Preface 1
1. Concepts and Architecture 3
1.1. Components 3
1.2. Request flow 3
1.3. Deployment models 4
1.4. Capabilities and limitations 4
2. Installation and Deployment 5
2.1. Requirements 5
2.2. Client-managed container with stdio 5
2.3. Standalone HTTP server 5
3. Server Configuration 7
3.1. Uyuni connection 7
3.2. Transport and service 7
3.3. Tools and logging 8
3.4. System bootstrap 8
3.5. Example read-only configuration 8
4. MCP Client Configuration 9
4.1. Configure a stdio client 9
4.2. Configure an HTTP client 9
4.3. Verify the integration 9
4.4. Use tools efficiently 10
5. Security 11
5.1. Security model 11
5.2. OAuth configuration 11
5.3. Write tool controls 12
5.4. Hardening the deployment 12
5.5. AI and prompt security 13
6. Tool Reference 14
6.1. Systems and events 14
6.2. Updates and CVEs 14
6.3. Actions, activation keys, and groups 15
6.4. State-changing tools 15
6.5. Pagination controls 16
7. Troubleshooting 17
7.1. Startup problems 17
7.2. Authentication and authorization 17
7.3. Connectivity and certificates 18
7.4. Tool-call problems 18
8. GNU Free Documentation License 19

장 1. Concepts and Architecture

This chapter introduces the components, request flow, deployment models, and capabilities of the Uyuni MCP Server.

The Model Context Protocol (MCP) is an open protocol that allows an AI application to discover and call tools exposed by an external server. The Uyuni MCP Server implements MCP tools that translate structured tool calls into Uyuni HTTP API requests.

The MCP Server integrates with the existing Uyuni authorization model. The authenticated account determines which content is visible and which operations are permitted.

1.1. Components

An MCP integration contains the following components:

MCP host

The AI application in which the user works.

MCP client

The protocol client maintained by the host. It discovers tools, sends tool calls, displays progress and results, and can request user approval.

Uyuni MCP Server

The process that exposes tools over stdio or streamable HTTP and calls the Uyuni API.

Uyuni Server

The managed infrastructure service and authorization boundary.

Identity provider

An optional OAuth 2.0/OpenID Connect provider used for a shared HTTP deployment.

1.2. Request flow

For a read-only request, the flow is:

1. The user asks the AI assistant to retrieve information from Uyuni.
2. The AI assistant selects an MCP tool and supplies structured arguments.
3. The MCP client sends the tool call to the MCP Server.
4. The MCP Server authenticates to Uyuni, calls the corresponding API, and normalizes the response.
5. The MCP client returns the structured result to the AI assistant.

State-changing tools add an approval step when the MCP client supports MCP elicitation. For more information, see Write tool controls.

1.3. Deployment models

Model	Use case	Characteristics
Client-managed stdio	One user or one local AI application	The MCP client starts one server process. Communication remains on stdin and stdout. Uyuni credentials are normally supplied to that process.
Standalone HTTP	A persistent service shared by multiple users or clients	Clients connect to **/mcp** over streamable HTTP. OAuth is strongly recommended. Production deployments should expose the endpoint only over TLS.

Use client-managed stdio unless the MCP Server must be shared or centrally operated. The smaller network surface and process lifetime make stdio easier to secure.

1.4. Capabilities and limitations

The server exposes MCP tools for systems, updates, CVEs, actions, activation keys, and system groups. Write tools are disabled by default. The server does not expose MCP resources or prompts.

Tool results can be paginated to reduce context usage. For fleet-wide questions, use summary tools first and request detailed update or CVE data only when required.



An AI-generated explanation is not an authoritative record of system state. Review the structured tool result and confirm important changes in the Uyuni Web UI, especially after scheduling updates, rebooting, bootstrapping, or removing systems.

장 2. Installation and Deployment

This chapter describes the requirements and procedures for deploying the Uyuni MCP Server as a client-managed stdio container or a standalone HTTP service.

The MCP Server is available as a container image from the SUSE Container Registry.

2.1. Requirements

Before deployment, ensure that:

- The deployment environment allows outbound HTTPS connections to the Uyuni Server.
- The Uyuni account used by the MCP Server has only the permissions required for the intended tools.
- A supported MCP client is installed.
- The MCP client supports streamable HTTP and OAuth when these features are required.

2.2. Client-managed container with stdio

In this model, the MCP client starts the container and communicates with it over stdio. No host port is published.

1. Create a protected environment file for the server configuration:

```
UYUNI_SERVER=https://server.example.com
UYUNI_USER=mcp-reader
UYUNI_PASS=replace-me
UYUNI_MCP_SSL_VERIFY=true
UYUNI_MCP_WRITE_TOOLS_ENABLED=false
```

   Restrict access to this file because it contains credentials. See Server Configuration for all settings.

2. Configure the MCP client to run:

```
podman run -i --rm \
  --env-file /absolute/path/to/mcp.env \
  registry.suse.com/suse/agentic/mcp/multi-linux-manager:<version>
```

   Replace **<version>** with a tested release tag. Pin a release tag instead of using **latest** in a production configuration.

Continue with stdio client configuration.

2.3. Standalone HTTP server

Use HTTP mode when a persistent service must support multiple clients.

1. Create a protected environment file for the server configuration:

```
UYUNI_SERVER=https://server.example.com
UYUNI_MCP_SSL_VERIFY=true
UYUNI_MCP_WRITE_TOOLS_ENABLED=false
UYUNI_MCP_TRANSPORT=http
UYUNI_MCP_HOST=0.0.0.0
UYUNI_MCP_PORT=8000
UYUNI_MCP_PUBLIC_URL=https://mcp.example.com
UYUNI_AUTH_SERVER=https://idp.example.com/realms/mcp
```

   When **UYUNI_AUTH_SERVER** is set, the MCP Server uses each client's OAuth token to authenticate to the Uyuni API. In this configuration, **UYUNI_USER** and **UYUNI_PASS** are not required.

2. Start the container:

```
podman run --rm \
  --env-file /absolute/path/to/mcp.env \
  -p 8000:8000 \
  registry.suse.com/suse/agentic/mcp/multi-linux-manager:<version>
```

3. Configure TLS termination for https://mcp.example.com and route requests to port **8000**.
4. Connect the MCP client to:

```
https://mcp.example.com/mcp
```

The application serves plain HTTP itself. Do not expose it directly to an untrusted network.



If **UYUNI_AUTH_SERVER** is not set, HTTP requests to the MCP Server are unauthenticated. Only omit authentication for temporary testing on an isolated interface or network.

Complete the OAuth configuration described in OAuth configuration before allowing user access.

장 3. Server Configuration

This chapter describes the environment variables for configuring the Uyuni connection, transport, tools, logging, and system bootstrap.

3.1. Uyuni connection

Variable	Required	Default	Description
UYUNI_SERVER	Yes	-	Base URL of the Uyuni Server. If the scheme is omitted, **https** is used. A path, query, or fragment is not allowed.
UYUNI_USER	For basic auth	-	Uyuni user name. Use with **UYUNI_PASS** when **UYUNI_AUTH_SERVER** is not set.
UYUNI_PASS	For basic auth	-	Password for **UYUNI_USER**.
UYUNI_MCP_SSL_VERIFY	No	**true**	Verifies the TLS certificate presented by the Uyuni Server. Values **false**, **0**, and **no** disable verification.
UYUNI_MCP_TIMEOUT	No	**30**	HTTP request timeout in seconds. The connection timeout remains 10 seconds.



Do not disable certificate verification in production. Install the issuing CA certificate in the container or host trust store instead.

3.2. Transport and service

Variable	Required	Default	Description
UYUNI_MCP_TRANSPORT	No	**stdio**	Transport mode. Valid values are **stdio** and **http**.
UYUNI_MCP_HOST	HTTP only	**127.0.0.1**	Address on which the HTTP server listens.
UYUNI_MCP_PORT	HTTP only	**8000**	Port on which the HTTP server listens.
UYUNI_MCP_PUBLIC_URL	OAuth HTTP deployment	Derived from host and port	External base URL advertised to OAuth-capable clients. Set the client-facing URL, without **/mcp**.
UYUNI_AUTH_SERVER	Recommended for HTTP	-	OpenID Connect issuer URL. A path is allowed. When omitted, the MCP HTTP endpoint does not authenticate clients.

3.3. Tools and logging

Variable	Required	Default	Description
UYUNI_MCP_WRITE_TOOLS_ENABLED	No	**false**	Registers tools that modify Uyuni. Values **true**, **1**, and **yes** enable them.
UYUNI_MCP_LOG_FILE_PATH	No	Console	Path to the server log file.
UYUNI_MCP_LOG_LEVEL	No	**INFO**	Logging level, for example **DEBUG**, **INFO**, **WARNING**, **ERROR**, or **CRITICAL**.



Changing **UYUNI_MCP_WRITE_TOOLS_ENABLED** requires a process restart because the server registers tools during startup.

3.4. System bootstrap

The **add_system** tool requires these additional variables:

Variable	Required	Description
UYUNI_SSH_PRIV_KEY	For **add_system**	Private SSH key used to bootstrap the target. Store the key as one line with each newline represented by the literal **\n** sequence.
UYUNI_SSH_PRIV_KEY_PASS	No	Passphrase for the private SSH key.



Do not place the key directly in an MCP client configuration that is synchronized or shared. Prefer a protected environment file or a secrets-management system.

3.5. Example read-only configuration

The following example uses basic authentication with **UYUNI_USER** and **UYUNI_PASS**.

```
UYUNI_SERVER=https://server.example.com
UYUNI_USER=mcp-reader
UYUNI_PASS=replace-me
UYUNI_MCP_SSL_VERIFY=true
UYUNI_MCP_TIMEOUT=30
UYUNI_MCP_TRANSPORT=stdio
UYUNI_MCP_WRITE_TOOLS_ENABLED=false
UYUNI_MCP_LOG_LEVEL=INFO
```

장 4. MCP Client Configuration

This chapter provides example stdio and HTTP client configurations and explains how to verify the integration.

MCP configuration formats differ across various clients. Use the examples in this chapter as a template and consult the client documentation for the exact property names and configuration file location.

4.1. Configure a stdio client

The following generic configuration starts a container for each MCP client session:

```
{
  "mcpServers": {
    "mcp-server-uyuni": {
      "command": "podman",
      "args": [
        "run",
        "-i",
        "--rm",
        "--env-file",
        "/absolute/path/to/mcp.env",
        "registry.suse.com/suse/agentic/mcp/multi-linux-manager:<version>"
      ]
    }
  }
}
```

The **-i** option is required because MCP uses the container's standard input and output. Use an absolute path to the environment file.

4.2. Configure an HTTP client

After deploying the standalone server, configure the client with its streamable HTTP endpoint:

```
{
  "mcpServers": {
    "mcp-server-uyuni": {
      "url": "https://mcp.example.com/mcp",
      "type": "http"
    }
  }
}
```

The client must be able to complete OAuth authorization against the issuer configured with **UYUNI_AUTH_SERVER**. The exact transport name in a client configuration can be **http** or **streamable-http**, depending on the client.

4.3. Verify the integration

Restart the MCP client after changing its configuration. Then make low-risk, read-only requests, for example:

- "List the first five active systems."
- "Find systems with **web** in the hostname."
- "List the first ten systems that need a reboot."

Confirm that the returned system names and IDs are aligned with the Uyuni Web UI.

4.4. Use tools efficiently

MCP tool output consumes AI context. When explicitly selecting tools, directing an assistant to use a particular tool, or implementing an agent workflow, apply these practices:

- Start with **summarize_system_updates** or **summarize_fleet_updates**.
- Use pagination parameters and follow **meta.next_offset**.
- Set **include_cves** or **include_updates** only when details are required.
- Use **find_systems_by_name** before a tool call when a hostname is ambiguous.
- Ask the assistant to show the target system ID, action ID, or erratum ID before a state-changing call.

장 5. Security

This chapter describes authentication, write-tool controls, deployment hardening, and security considerations for AI-assisted operations.

The MCP Server can query and modify managed infrastructure. Treat it as a privileged administrative interface.

5.1. Security model

The server supports two authentication paths:

Basic Uyuni authentication

In a stdio or trusted deployment, the server logs in with **UYUNI_USER** and **UYUNI_PASS**. All MCP clients that use that process receive the permissions of this account.

OAuth bearer token

In an authenticated HTTP deployment, the MCP Server validates the caller's token and forwards it to the Uyuni OIDC login endpoint. Uyuni then maps the token to an existing active user.

Use a dedicated, least-privileged Uyuni account for basic authentication. For a multi-user HTTP deployment, use OAuth so that actions retain individual user identity and permissions.

5.2. OAuth configuration

The MCP Server expects an OpenID Connect issuer. It retrieves signing keys from:

```
<UYUNI_AUTH_SERVER>/protocol/openid-connect/certs
```

Tokens accepted by the MCP Server must:

- Have an issuer that exactly matches **UYUNI_AUTH_SERVER**.
- Include the **mcp-server-uyuni** audience.
- Include the **mcp:read** scope.
- Include the **mcp:write** scope when write tools are enabled.

Configure the same identity provider on the Uyuni Server in **/etc/rhn/rhn.conf**:

```
web.oidc.enabled = true
web.oidc.idp.issuer = https://idp.example.com/realms/mcp
```

You can explicitly set the JSON Web Key Set path:

```
web.oidc.idp.jwks_path = /realms/mcp/protocol/openid-connect/certs
```

If the path is not set, Uyuni discovers it from the issuer's **/.well-known/openid-configuration** document.

The default Uyuni audience and username claim are:

```
web.oidc.jwt.audience = uyuni-server
web.oidc.jwt.username_claim = preferred_username
```

The token forwarded to Uyuni must include:

- The **uyuni-server** audience, in addition to the **mcp-server-uyuni** audience.
- A **sub** claim.
- A configured username claim whose value matches an existing active Uyuni user.

Restart the relevant Uyuni services after changing **rhn.conf**.

5.3. Write tool controls

State-changing tools are not registered unless **UYUNI_MCP_WRITE_TOOLS_ENABLED=true**.

When write tools are enabled, the server asks for confirmation before a change if the MCP client advertises the MCP elicitation capability. Clients without elicitation support do not receive this server-side confirmation. Such clients are responsible for their own approval user experience.



Do not rely on elicitation as the only authorization control. Use Uyuni role-based access control permissions, OAuth scopes, network restrictions, and client-side tool approval policies.

Enable write tools only for a deployment where they are required. For high-risk operations:

- Resolve and verify the target system ID.
- Review the generated arguments before approval.
- Apply changes to a small scope first.
- Verify the resulting scheduled action in the Uyuni Web UI.

Using the **remove_system** tool is a destructive operation. When **cleanup=true**, the tool requests forced deletion and cleanup of the system.

5.4. Hardening the deployment

You can harden the MCP Server deployment by applying the following controls:

- Provide TLS termination for HTTP deployments.
- Never expose unauthenticated HTTP mode to an untrusted network.
- Keep **UYUNI_MCP_SSL_VERIFY=true**.

- Store passwords, OAuth configuration, and SSH private keys in a secrets-management system or protected environment file.
- Pin the MCP Server container to a reviewed release.
- Use **INFO** or **WARNING** logging in production and protect log files from unauthorized access.
- Restrict network access from the MCP Server to the required Uyuni and identity-provider endpoints.

5.5. AI and prompt security

An AI assistant can use text from prompts and external content when deciding whether and how to call a tool. This content might contain misleading or malicious instructions. Treat any tool call based on untrusted content as untrusted until you review it.

Configure the MCP host to require explicit approval for state-changing tools. Before approval, verify the tool name, target IDs, update or action IDs, and all arguments. Do not provide secrets in prompts. Do not ask the model to reproduce environment variables, access tokens, passwords, or private keys.

장 6. Tool Reference

This chapter provides a reference to the tools exposed by the Uyuni MCP Server, including their arguments, results, and effects.

The Uyuni MCP Server exposes read-only tools by default. Tools in the state-changing section are available only when **UYUNI_MCP_WRITE_TOOLS_ENABLED=true**.

Most tools that accept **system_identifier** support a system name or system ID. If a name is not found or is ambiguous, call **find_systems_by_name** and use the returned ID.

6.1. Systems and events

Tool	Arguments	Result
list_systems	**limit=50**, **offset=0**	Paged active systems. **limit** is capped at 500.
get_system_details	**system_identifier**	System ID, name, last boot time, UUID, CPU, network, and installed products.
get_system_event_history	**system_identifier**, **offset=0**, **limit=10**, optional **earliest_date**	Newest-first event history.
get_system_event_details	**system_identifier**, **event_id**	Status, timestamps, result fields, and additional information for one event.
find_systems_by_name	**name**, **limit=50**, **offset=0**	Paged hostname matches.
find_systems_by_ip	**ip_address**, **limit=50**, **offset=0**	Paged systems matching an IP address.
list_systems_needing_reboot	**limit=50**, **offset=0**	Paged systems requiring a reboot. **limit** is capped at 500.

6.2. Updates and CVEs

The supported advisory type values are **Security Advisory**, **Product Enhancement Advisory**, and **Bug Fix Advisory**.

Tool	Arguments	Result
get_system_updates	**system_identifier**	Compact pending update list and counts. CVE expansion is omitted.
summarize_system_updates	**system_identifier**, optional **advisory_types**	Pending and queued update counts without update details.
query_system_updates	**system_identifier**, **limit=25**, **offset=0**, **include_cves=false**, optional **advisory_types**	Paged pending updates. A positive **limit** is capped at 200; a non-positive value returns counts without items.

Tool	Arguments	Result
check_all_systems_for_updates	**include_updates=false**, **include_cves=false**, **system_limit=25**, **system_offset=0**, **updates_per_system=10**	Paged systems with pending updates. **system_limit** is capped at 200.
summarize_fleet_updates	**system_limit=25**, **system_offset=0**	Paged systems and update counts without update details. **system_limit** is capped at 200.
list_systems_needing_update_for_cve	**cve_identifier**, **limit=50**, **offset=0**	Paged systems affected by the specified CVE.
get_unscheduled_errata	**system_id**	Applicable patches that are not scheduled. **system_id** must be a numeric string.

6.3. Actions, activation keys, and groups

Tool	Arguments	Result
list_all_scheduled_actions	**limit=50**, **offset=0**, optional **action_types**, optional **scheduler**	Paged actions after exact-match filtering. **limit** is capped at 500. The metadata lists observed action types and schedulers for subsequent filters.
list_activation_keys	None	Activation keys available to the current user.
list_system_groups	None	Group ID, name, description, and system count.
list_group_systems	**group_name**	System IDs and names in the specified group.

6.4. State-changing tools



These tools change managed infrastructure. Review all arguments and the intended targets before approving a call.

Tool	Arguments	Effect
schedule_pending_updates_to_system	**system_identifier**	Schedules all pending updates for one system.
schedule_specific_update	**system_identifier**, **errata_id**	Schedules one patch for one system.
add_system	**host**, optional **activation_key**, **ssh_port=22**, **ssh_user=root**, **proxy_id**, **salt_ssh=false**	Bootstraps and registers a system over SSH. Requires **UYUNI_SSH_PRIV_KEY**. If the client supports elicitation, a missing activation key can be requested interactively.

Tool	Arguments	Effect
remove_system	**system_identifier**, **cleanup=true**	Removes a system. **cleanup=true** requests forced deletion and cleanup; **false** requests no cleanup.
schedule_system_reboot	**system_identifier**	Schedules an immediate reboot.
cancel_action	**action_id**	Cancels the specified scheduled action.
create_system_group	**name**, optional **description**	Creates a system group.
add_systems_to_group	**group_name**, **system_identifiers**	Adds the specified systems to a group.
remove_systems_from_group	**group_name**, **system_identifiers**	Removes the specified systems from a group.

The **add_system** API can continue processing after the MCP Server request timeout. When the tool reports that the addition process has started, check the system list later rather than immediately repeating the call.

6.5. Pagination controls

Tools that can return large result sets support pagination with **limit** and **offset** arguments. Paginated responses include a **meta** object that describes the current page and provides **next_offset** when more results are available. Pagination limits response size, reduces latency, and lowers AI context usage.

장 7. Troubleshooting

This chapter describes common problems that can occur when configuring or operating the Uyuni MCP Server and provides guidance for resolving them.

Start troubleshooting with a read-only tool such as **list_systems**. Set **UYUNI_MCP_LOG_LEVEL=DEBUG** temporarily when the default log level does not provide enough information. Debug logs can contain sensitive operational data; protect and remove them according to your logging policy.

7.1. Startup problems

Message or symptom	Resolution
Missing required environment variables: UYUNI_SERVER	Set **UYUNI_SERVER** in the environment visible to the MCP Server process. For a client-managed container, verify the absolute path supplied to **--env-file**.
UYUNI_SERVER must use http or https and include a host	Use a URL such as https://server.example.com.
UYUNI_SERVER must not include a path	Remove paths such as **/rhn** and use only the server base URL.
UYUNI_MCP_TRANSPORT must be either 'stdio' or 'http'	Set the value to exactly **stdio** or **http**.
The stdio server disconnects immediately	Run the configured command in a terminal and inspect stderr. Verify the image tag, environment-file path, **-i** container option, and Uyuni URL.
The HTTP client receives a connection error	Verify the bind address, published port, external route, and that the client URL ends in **/mcp**.

7.2. Authentication and authorization

Message or symptom	Resolution
HTTP 401 or 403 during Uyuni login	For basic authentication, verify **UYUNI_USER**, **UYUNI_PASS**, account status, and permissions. For OAuth, verify issuer, audiences, scopes, username claim, and the matching active Uyuni user.
OAuth discovery or token validation fails	Ensure **UYUNI_MCP_PUBLIC_URL** is the externally reachable HTTPS base URL. Confirm that **UYUNI_AUTH_SERVER** exactly matches the token **iss** claim and that its signing-key endpoint is reachable by the MCP Server.
MCP authentication succeeds but Uyuni OIDC login fails	Ensure the token includes both **mcp-server-uyuni** and **uyuni-server** audiences, includes **sub**, and contains the configured username claim.

Message or symptom	Resolution
A read tool is available but a write tool is missing	Set **UYUNI_MCP_WRITE_TOOLS_ENABLED=true** and restart the MCP Server. In OAuth mode, obtain a token with both **mcp:read** and **mcp:write**.
The user can call a tool but the operation is denied	Grant only the required Uyuni permissions to the user. MCP availability does not override Uyuni authorization.

7.3. Connectivity and certificates

A network error or timeout indicates that the MCP Server could not complete a request to Uyuni.

1. Resolve **UYUNI_SERVER** from the MCP Server host or container.
2. Verify TCP and TLS connectivity to the configured port.
3. Confirm that proxies and firewalls allow the connection.
4. Verify that the server certificate is valid for the configured hostname and that its CA is trusted.
5. Increase **UYUNI_MCP_TIMEOUT** only after checking server and network health.

Do not use **UYUNI_MCP_SSL_VERIFY=false** as a permanent certificate fix.

7.4. Tool-call problems

Message or symptom	Resolution
A system name is not found or resolves ambiguously	Call **find_systems_by_name** and pass the numeric **system_id** to the next tool.
Invalid system_id ⋯ Provide system_id as a numeric string	The **get_unscheduled_errata** tool does not accept a hostname. Supply the system ID as digits.
You need to provide an activation key	The MCP client does not support elicitation. Pass **activation_key** explicitly to **add_system**.
UYUNI_SSH_PRIV_KEY environment variable is not set	Set the complete private key with literal **\n** separators in the MCP Server environment and restart the process.
System addition reports that the process started	The bootstrap request exceeded the request timeout but can still be running on Uyuni. Check the system list and server logs before retrying.
An operation returns **Operation cancelled**	The elicitation request was declined, cancelled, or not explicitly approved. Review the arguments and invoke the tool again only if the change is intended.

장 8. GNU Free Documentation License

Copyright © 2000, 2001, 2002 Free Software Foundation, Inc. 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA. Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

0. PREAMBLE

The purpose of this License is to make a manual, textbook, or other functional and useful document "free" in the sense of freedom: to assure everyone the effective freedom to copy and redistribute it, with or without modifying it, either commercially or noncommercially. Secondarily, this License preserves for the author and publisher a way to get credit for their work, while not being considered responsible for modifications made by others.

This License is a kind of "copyleft", which means that derivative works of the document must themselves be free in the same sense. It complements the GNU General Public License, which is a copyleft license designed for free software.

We have designed this License in order to use it for manuals for free software, because free software needs free documentation: a free program should come with manuals providing the same freedoms that the software does. But this License is not limited to software manuals; it can be used for any textual work, regardless of subject matter or whether it is published as a printed book. We recommend this License principally for works whose purpose is instruction or reference.

1. APPLICABILITY AND DEFINITIONS

This License applies to any manual or other work, in any medium, that contains a notice placed by the copyright holder saying it can be distributed under the terms of this License. Such a notice grants a world-wide, royalty-free license, unlimited in duration, to use that work under the conditions stated herein. The "Document", below, refers to any such manual or work. Any member of the public is a licensee, and is addressed as "you". You accept the license if you copy, modify or distribute the work in a way requiring permission under copyright law.

A "Modified Version" of the Document means any work containing the Document or a portion of it, either copied verbatim, or with modifications and/or translated into another language.

A "Secondary Section" is a named appendix or a front-matter section of the Document that deals exclusively with the relationship of the publishers or authors of the Document to the Document’s overall subject (or to related matters) and contains nothing that could fall directly within that overall subject. (Thus, if the Document is in part a textbook of mathematics, a Secondary Section may not explain any mathematics.) The relationship could be a matter of historical connection with the subject or with related matters, or of legal, commercial, philosophical, ethical or political position regarding them.

The "Invariant Sections" are certain Secondary Sections whose titles are designated, as being those of Invariant Sections, in the notice that says that the Document is released under this License. If a section does not fit the above definition of Secondary then it is not allowed to be designated as Invariant. The Document may contain zero Invariant Sections. If the Document does not identify any Invariant Sections then there are none.

The "Cover Texts" are certain short passages of text that are listed, as Front-Cover Texts or Back-Cover Texts, in the notice that says that the Document is released under this License. A Front-Cover Text may be at most 5 words, and a Back-Cover Text may be at most 25 words.

A "Transparent" copy of the Document means a machine-readable copy, represented in a format whose specification is available to the general public, that is suitable for revising the document straightforwardly with generic text editors or (for images composed of pixels) generic paint programs or (for drawings) some widely available drawing editor, and that is suitable for input to text formatters or for automatic translation to a variety of formats suitable for input to text formatters. A copy made in an otherwise Transparent file format whose markup, or absence of markup, has been arranged to thwart or discourage subsequent modification by readers is not Transparent. An image format is not Transparent if used for any substantial amount of text. A copy that is not "Transparent" is called "Opaque".

Examples of suitable formats for Transparent copies include plain ASCII without markup, Texinfo input format, LaTeX input format, SGML or XML using a publicly available DTD, and standard-conforming simple HTML, PostScript or PDF designed for human modification. Examples of transparent image formats include PNG, XCF and JPG. Opaque formats include proprietary formats that can be read and edited only by proprietary word processors, SGML or XML for which the DTD and/or processing tools are not generally available, and the machine-generated HTML, PostScript or PDF produced by some word processors for output purposes only.

The "Title Page" means, for a printed book, the title page itself, plus such following pages as are needed to hold, legibly, the material this License requires to appear in the title page. For works in formats which do not have any title page as such, "Title Page" means the text near the most prominent appearance of the work’s title, preceding the beginning of the body of the text.

A section "Entitled XYZ" means a named subunit of the Document whose title either is precisely XYZ or contains XYZ in parentheses following text that translates XYZ in another language. (Here XYZ stands for a specific section name mentioned below, such as "Acknowledgements", "Dedications", "Endorsements", or "History".) To "Preserve the Title" of such a section when you modify the Document means that it remains a section "Entitled XYZ" according to this definition.

The Document may include Warranty Disclaimers next to the notice which states that this License applies to the Document. These Warranty Disclaimers are considered to be included by reference in this License, but only as regards disclaiming warranties: any other implication that these Warranty Disclaimers may have is void and has no effect on the meaning of this License.

2. VERBATIM COPYING

You may copy and distribute the Document in any medium, either commercially or noncommercially, provided that this License, the copyright notices, and the license notice saying this License applies to the Document are reproduced in all copies, and that you add no other conditions whatsoever to those of this License. You may not use technical measures to obstruct or control the reading or further copying of the copies you make or distribute. However, you may accept compensation in exchange for copies. If you distribute a large enough number of copies you must also follow the conditions in section 3.

You may also lend copies, under the same conditions stated above, and you may publicly display copies.

3. COPYING IN QUANTITY

If you publish printed copies (or copies in media that commonly have printed covers) of the Document, numbering more than 100, and the Document's license notice requires Cover Texts, you must enclose the copies in covers that carry, clearly and legibly, all these Cover Texts: Front-Cover Texts on the front cover, and Back-Cover Texts on the back cover. Both covers must also clearly and legibly identify you as the publisher of these copies. The front cover must present the full title with all words of the title equally prominent and visible. You may add other material on the covers in addition. Copying with changes limited to the covers, as long as they preserve the title of the Document and satisfy these conditions, can be treated as verbatim copying in other respects.

If the required texts for either cover are too voluminous to fit legibly, you should put the first ones listed (as many as fit reasonably) on the actual cover, and continue the rest onto adjacent pages.

If you publish or distribute Opaque copies of the Document numbering more than 100, you must either include a machine-readable Transparent copy along with each Opaque copy, or state in or with each Opaque copy a computer-network location from which the general network-using public has access to download using public-standard network protocols a complete Transparent copy of the Document, free of added material. If you use the latter option, you must take reasonably prudent steps, when you begin distribution of Opaque copies in quantity, to ensure that this Transparent copy will remain thus accessible at the stated location until at least one year after the last time you distribute an Opaque copy (directly or through your agents or retailers) of that edition to the public.

It is requested, but not required, that you contact the authors of the Document well before redistributing any large number of copies, to give them a chance to provide you with an updated version of the Document.

4. MODIFICATIONS

You may copy and distribute a Modified Version of the Document under the conditions of sections 2 and 3 above, provided that you release the Modified Version under precisely this License, with the Modified Version filling the role of the Document, thus licensing distribution and modification of the Modified Version to whoever possesses a copy of it. In addition, you must do these things in the Modified Version:

A. Use in the Title Page (and on the covers, if any) a title distinct from that of the Document, and from those of previous versions (which should, if there were any, be listed in the History section of the Document). You may use the same title as a previous version if the original publisher of that version gives permission.

B. List on the Title Page, as authors, one or more persons or entities responsible for authorship of the modifications in the Modified Version, together with at least five of the principal authors of the Document (all of its principal authors, if it has fewer than five), unless they release you from this requirement.

C. State on the Title page the name of the publisher of the Modified Version, as the publisher.

D. Preserve all the copyright notices of the Document.

E. Add an appropriate copyright notice for your modifications adjacent to the other copyright notices.

F. Include, immediately after the copyright notices, a license notice giving the public permission to use the Modified Version under the terms of this License, in the form shown in the Addendum

below.

G. Preserve in that license notice the full lists of Invariant Sections and required Cover Texts given in the Document’s license notice.

H. Include an unaltered copy of this License.

I. Preserve the section Entitled "History", Preserve its Title, and add to it an item stating at least the title, year, new authors, and publisher of the Modified Version as given on the Title Page. If there is no section Entitled "History" in the Document, create one stating the title, year, authors, and publisher of the Document as given on its Title Page, then add an item describing the Modified Version as stated in the previous sentence.

J. Preserve the network location, if any, given in the Document for public access to a Transparent copy of the Document, and likewise the network locations given in the Document for previous versions it was based on. These may be placed in the "History" section. You may omit a network location for a work that was published at least four years before the Document itself, or if the original publisher of the version it refers to gives permission.

K. For any section Entitled "Acknowledgements" or "Dedications", Preserve the Title of the section, and preserve in the section all the substance and tone of each of the contributor acknowledgements and/or dedications given therein.

L. Preserve all the Invariant Sections of the Document, unaltered in their text and in their titles. Section numbers or the equivalent are not considered part of the section titles.

M. Delete any section Entitled "Endorsements". Such a section may not be included in the Modified Version.

N. Do not retitle any existing section to be Entitled "Endorsements" or to conflict in title with any Invariant Section.

O. Preserve any Warranty Disclaimers.

If the Modified Version includes new front-matter sections or appendices that qualify as Secondary Sections and contain no material copied from the Document, you may at your option designate some or all of these sections as invariant. To do this, add their titles to the list of Invariant Sections in the Modified Version’s license notice. These titles must be distinct from any other section titles.

You may add a section Entitled "Endorsements", provided it contains nothing but endorsements of your Modified Version by various parties—for example, statements of peer review or that the text has been approved by an organization as the authoritative definition of a standard.

You may add a passage of up to five words as a Front-Cover Text, and a passage of up to 25 words as a Back-Cover Text, to the end of the list of Cover Texts in the Modified Version. Only one passage of Front-Cover Text and one of Back-Cover Text may be added by (or through arrangements made by) any one entity. If the Document already includes a cover text for the same cover, previously added by you or by arrangement made by the same entity you are acting on behalf of, you may not add another; but you may replace the old one, on explicit permission from the previous publisher that added the old one.

The author(s) and publisher(s) of the Document do not by this License give permission to use their names for publicity for or to assert or imply endorsement of any Modified Version.

5. COMBINING DOCUMENTS

You may combine the Document with other documents released under this License, under the terms defined in section 4 above for modified versions, provided that you include in the combination all of the Invariant Sections of all of the original documents, unmodified, and list them all as Invariant Sections of your combined work in its license notice, and that you preserve all their Warranty Disclaimers.

The combined work need only contain one copy of this License, and multiple identical Invariant Sections may be replaced with a single copy. If there are multiple Invariant Sections with the same name but different contents, make the title of each such section unique by adding at the end of it, in parentheses, the name of the original author or publisher of that section if known, or else a unique number. Make the same adjustment to the section titles in the list of Invariant Sections in the license notice of the combined work.

In the combination, you must combine any sections Entitled "History" in the various original documents, forming one section Entitled "History"; likewise combine any sections Entitled "Acknowledgements", and any sections Entitled "Dedications". You must delete all sections Entitled "Endorsements".

6. COLLECTIONS OF DOCUMENTS

You may make a collection consisting of the Document and other documents released under this License, and replace the individual copies of this License in the various documents with a single copy that is included in the collection, provided that you follow the rules of this License for verbatim copying of each of the documents in all other respects.

You may extract a single document from such a collection, and distribute it individually under this License, provided you insert a copy of this License into the extracted document, and follow this License in all other respects regarding verbatim copying of that document.

7. AGGREGATION WITH INDEPENDENT WORKS

A compilation of the Document or its derivatives with other separate and independent documents or works, in or on a volume of a storage or distribution medium, is called an "aggregate" if the copyright resulting from the compilation is not used to limit the legal rights of the compilation’s users beyond what the individual works permit. When the Document is included in an aggregate, this License does not apply to the other works in the aggregate which are not themselves derivative works of the Document.

If the Cover Text requirement of section 3 is applicable to these copies of the Document, then if the Document is less than one half of the entire aggregate, the Document’s Cover Texts may be placed on covers that bracket the Document within the aggregate, or the electronic equivalent of covers if the Document is in electronic form. Otherwise they must appear on printed covers that bracket the whole aggregate.

8. TRANSLATION

Translation is considered a kind of modification, so you may distribute translations of the Document under the terms of section 4. Replacing Invariant Sections with translations requires special permission from their copyright holders, but you may include translations of some or all Invariant Sections in addition to the original versions of these Invariant Sections. You may include a translation of this

License, and all the license notices in the Document, and any Warranty Disclaimers, provided that you also include the original English version of this License and the original versions of those notices and disclaimers. In case of a disagreement between the translation and the original version of this License or a notice or disclaimer, the original version will prevail.

If a section in the Document is Entitled "Acknowledgements", "Dedications", or "History", the requirement (section 4) to Preserve its Title (section 1) will typically require changing the actual title.

9. TERMINATION

You may not copy, modify, sublicense, or distribute the Document except as expressly provided for under this License. Any other attempt to copy, modify, sublicense or distribute the Document is void, and will automatically terminate your rights under this License. However, parties who have received copies, or rights, from you under this License will not have their licenses terminated so long as such parties remain in full compliance.

10. FUTURE REVISIONS OF THIS LICENSE

The Free Software Foundation may publish new, revised versions of the GNU Free Documentation License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns. See http://www.gnu.org/copyleft/.

Each version of the License is given a distinguishing version number. If the Document specifies that a particular numbered version of this License "or any later version" applies to it, you have the option of following the terms and conditions either of that specified version or of any later version that has been published (not as a draft) by the Free Software Foundation. If the Document does not specify a version number of this License, you may choose any version ever published (not as a draft) by the Free Software Foundation.

ADDENDUM: How to use this License for your documents

Copyright (c) YEAR YOUR NAME. Permission is granted to copy, distribute and/or modify this document under the terms of the GNU Free Documentation License, Version 1.2 or any later version published by the Free Software Foundation; with no Invariant Sections, no Front-Cover Texts, and no Back-Cover Texts. A copy of the license is included in the section entitled "GNU Free Documentation License".